



INDIAN INSTITUTE OF TECHNOLOGY
KHARAGPUR

Stamp / Signature of the Invigilator

EXAMINATION (Mid Semester)

SEMESTER (Spring)

Roll Number

Section

Name

Subject Number

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Subject Name

Computer Networks

Department / Center of the Student

Additional sheets

Important Instructions and Guidelines for Students

1. You must occupy your seat as per the Examination Schedule/Sitting Plan.
2. Do not keep mobile phones or any similar electronic gadgets with you even in the switched off mode.
3. Loose papers, class notes, books or any such materials must not be in your possession, even if they are irrelevant to the subject you are taking examination.
4. Data book, codes, graph papers, relevant standard tables/charts or any other materials are allowed only when instructed by the paper-setter.
5. Use of instrument box, pencil box and non-programmable calculator is allowed during the examination. However, exchange of these items or any other papers (including question papers) is not permitted.
6. Write on both sides of the answer script and do not tear off any page. **Use last page(s) of the answer script for rough work.** Report to the invigilator if the answer script has torn or distorted page(s).
7. It is your responsibility to ensure that you have signed the Attendance Sheet. Keep your Admit Card/Identity Card on the desk for checking by the invigilator.
8. You may leave the examination hall for wash room or for drinking water for a very short period. Record your absence from the Examination Hall in the register provided. Smoking and the consumption of any kind of beverages are strictly prohibited inside the Examination Hall.
9. Do not leave the Examination Hall without submitting your answer script to the invigilator. **In any case, you are not allowed to take away the answer script with you.** After the completion of the examination, do not leave the seat until the invigilators collect all the answer scripts.
10. During the examination, either inside or outside the Examination Hall, gathering information from any kind of sources or exchanging information with others or any such attempt will be treated as 'unfair means'. Do not adopt unfair means and do not indulge in unseemly behavior.

Violation of any of the above instructions may lead to severe punishment.

Signature of the Student

To be filled in by the examiner

Question Number

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Total

Marks Obtained

Marks obtained (in words)

Signature of the Examiner

Signature of the Scrutineer

Note:

- (i) There are TWO parts of this question paper – PART A and PART B. There are THREE questions in PART A and FOUR questions in PART B. Answer all the questions. The answers should be precise and to-the-point. Marks will be deducted for unnecessary texts.
 - (ii) Write down the assumptions clearly, if any. No clarifications will be given during the exam hours.
 - (iii) Maximum marks for the THREE questions in Part A: 20 marks, and the FOUR questions in Part B: 30 marks.
 - (iv) Write the answers in the boxes only. This question has 10 pages including the space for rough works. No additional sheets will be provided.
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PART A

1. (a) Host A is sending real-time voice to Host B over a packet-switched network. Host A converts analog voice to a digital 64 kbps bit stream on the fly, and then groups the bits into 56-Byte packets. As soon as Host B receives a packet, it converts the packet's bits to analog voice signal. All packets travel over three links, connected by two packet switches. If the propagation speed on all three links is 2.5×10^8 m/s, the transmission rates of all three links are 2.5 Mbps, the packet switch processing delay is 3 msec, the length of the first link is 5000 km, second link is 4000 km and the last link is 1000 km, what is the end-to-end delay for a packet to travel from Host A to Host B? Show all calculation steps. **[5 Marks]**

- (b) Your IP address is 100.100.100.100. A wireshark output indicates that you have received an HTTP packet on port 34262 from IP address 200.200.200.200 with a remaining TTL value of 35, and the application data captured is as follows:

```
HTTP/1.1 200 OK\r\n
Content-Length: 31\r\n
<html><p>Hello There!</p></html>
```

Using all of the above information, fill in the appropriate header fields in the table below. You do not need to populate all the fields in the headers; just figure out which of the above information will go to which headers in the respective layers of the protocol stack. **[3 Marks]**

Network Layer	Transport Layer	Application Layer

- (c) Consider an application that transmits data at a steady rate N bit per second and once started, it continues running for a relatively long period of time. Which among a packet-switched network or a circuit-switched network be more appropriate for this application? Why? **[2 Marks]**

2. You are preparing a website `www.example.com` that has four replicated servers in Delhi, Mumbai, Kolkata and Bengaluru. You can assume that the server at location `x` can be accessed via `x.example.com`. You want to use DNS as a load balancer by directing traffic to the nearby server. Every user, regardless of their location, attempts to connect to `www.example.com`. Show the series of steps (DNS queries and how the DNS server(s) handle them) that occur as the URL gets translated to the IP address of the optimal server. You can assume the DNS resolver to be recursive. **[5 Marks]**
[Bonus marks for considering the role of CNAME and A records and including in the answer.]

3. Alice is trying to send an email from `alice@abc.com` to `bob@xyz.com` using an email client configured with SMTP. Alice's email client connects to `mail.abc.com`, which further connects to `mail.xyz.com` to send the email. During this entire process: (a) Explain the sequence of commands and responses exchanged. (b) Describe the role of DNS in determining the correct mail servers for `abc.com` and `xyz.com`. **[3+2 = 5 Marks]**

PART B

4. Write whether the following statements are TRUE/FALSE with a short justification. No marks will be given if the justification is incorrect. **[(5 x 2) = 10 Marks]**

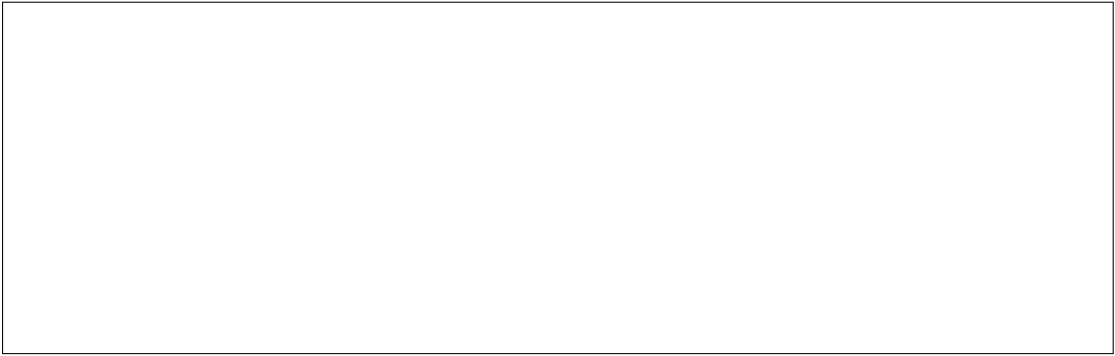
- i) For a connection oriented protocol running over a noisy communication network, the three-way handshaking during connection termination can ensure loss-free data transmission.

- ii) It is possible to implement a stop-and-wait ARQ with a 4 bit sequence number space.

- iii) AIMD-based congestion control algorithms can ensure absolute fairness among the contending flows.

- iv) Assume you are implementing a stream socket. You made five `send()` call on this socket, each call having 16 Kbytes of application data. TCP will always generate five segments out of these data and send the segments to the network layer.

- v) Nagle's Algorithm with delayed acknowledgment helps in reducing the queuing latency at the transport buffer.

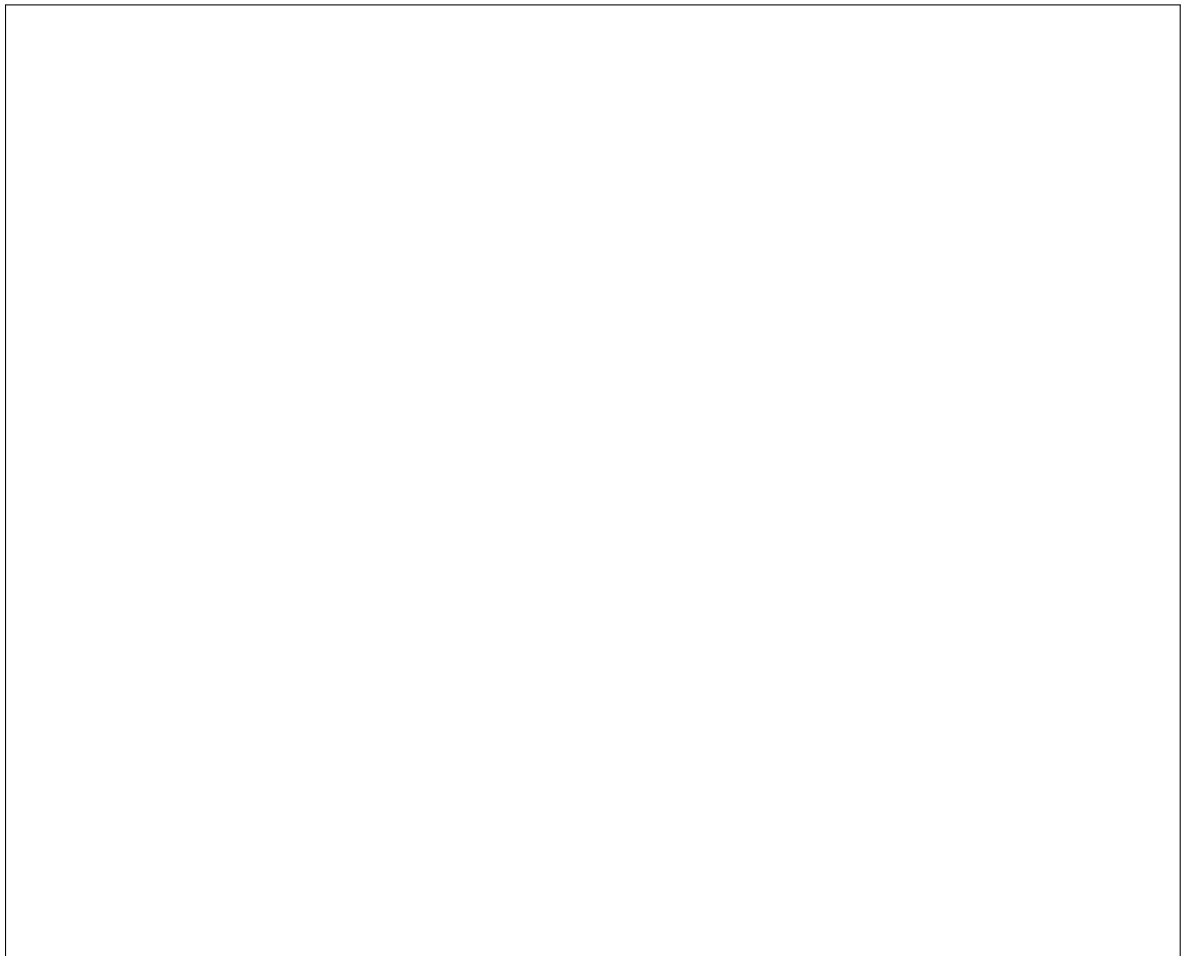


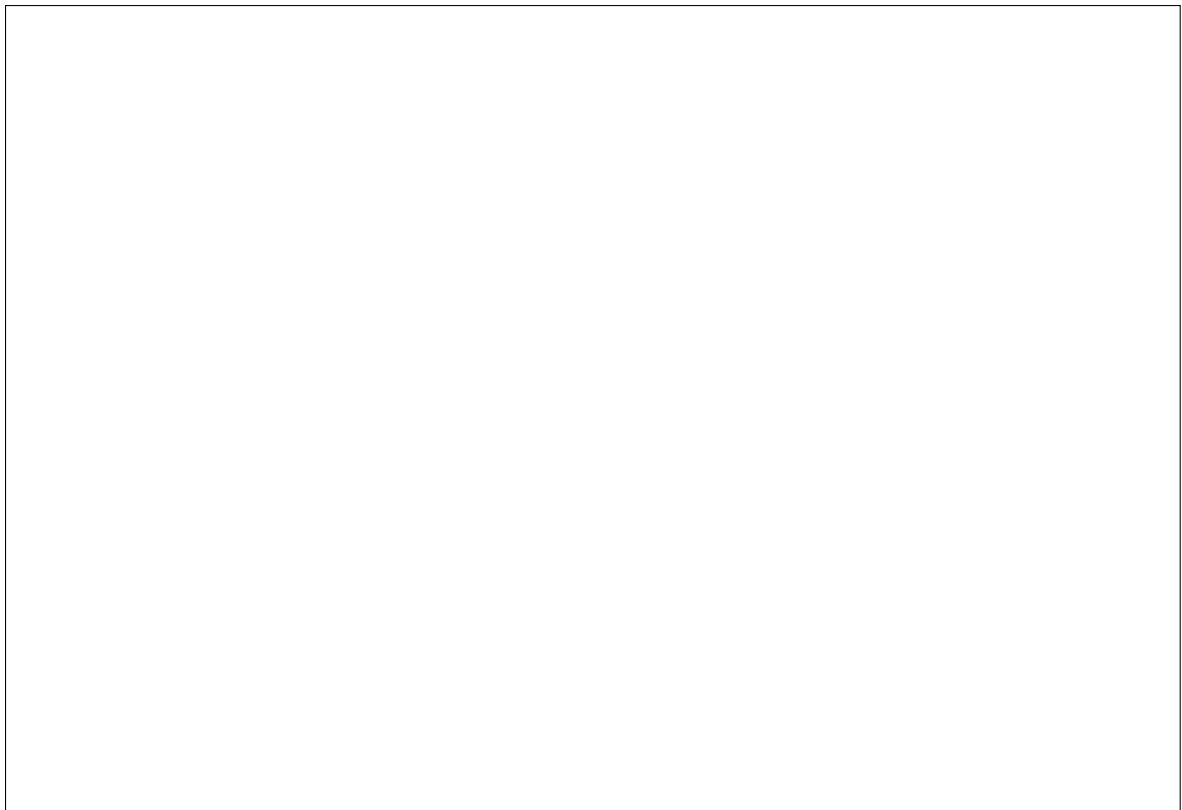
5. Consider a TCP sender that is using a 32-bit sequence number space and has a Maximum Segment Lifetime (MSL) of 60 seconds. The link between the sender and receiver has a bandwidth of 100 Mbps and a round-trip time (RTT) of 100 ms.

- (i) What should be the minimum transport buffer size at the TCP sender to fully utilize the network bandwidth?
- (ii) Compute the maximum sequence number space used during the MSL period.
- (iii) Determine the forbidden region size in the sequence number space.
- (iv) If the TCP sender transmits at full capacity, how long must it wait before safely wrapping around the sequence numbers without violating the forbidden region?

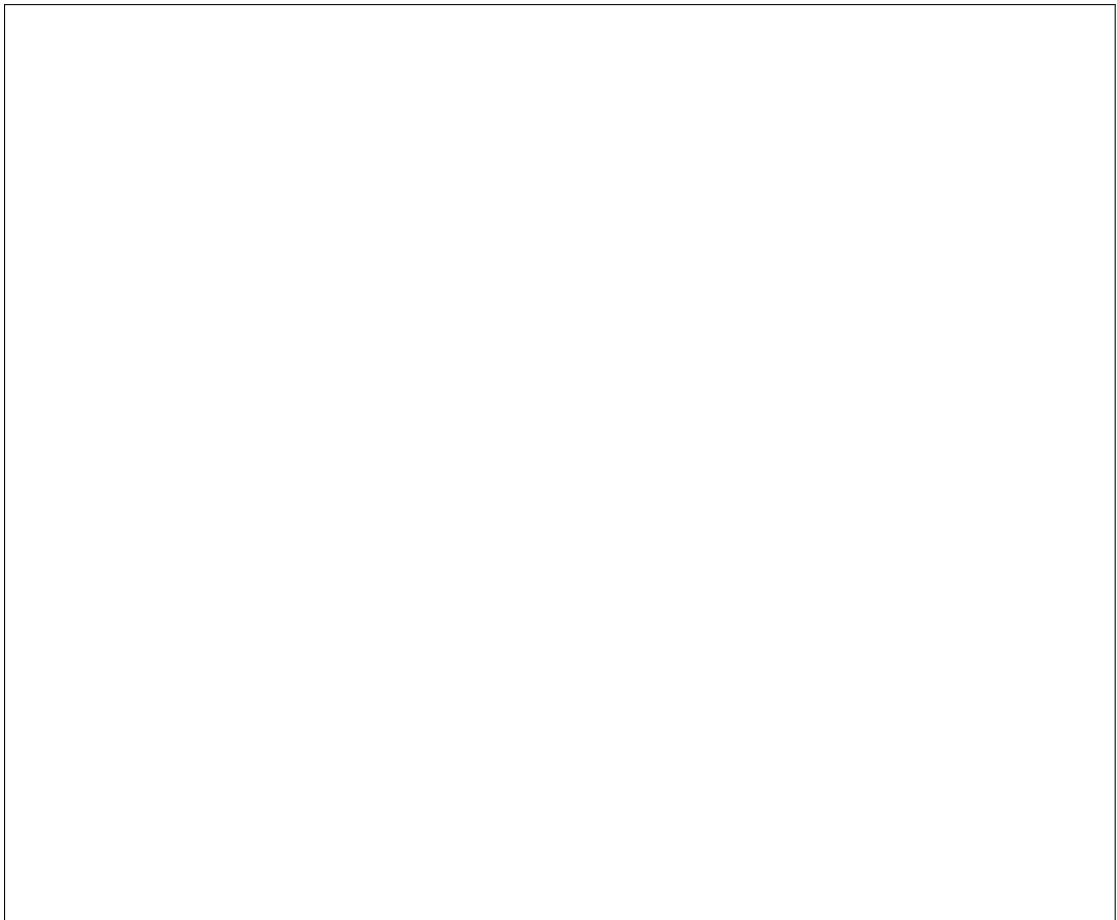
[Hint: TCP uses byte sequence number.]

[1+2+1+2 = 6 Marks]





6. (a) Draw the state machine that explains the sender side functionality of the Selective Repeat ARQ mechanism. You do not need to explain anything. Just draw the state transition diagram marking the input and output actions for each of the transitions. **[4 Marks]**



- (b) Consider a transport protocol that runs on a network having 1 Gbps bandwidth and 100ms RTT. The protocol implements Selective Repeat flow control algorithm with byte sequence numbers. What should the minimum number of bits to be used in the sequence number space to implement the protocol effectively? **[4 Marks]**

7. (a) Explain the issue with the silly window syndrome in TCP. How do we solve this? **[2+1 = 3 Marks]**

- (b) TCP detects network congestion in two ways – (a) retransmission timeout, and (b) duplicate acknowledgment. How does the TCP congestion control response for these two cases? Why does TCP response differently for these two cases? **[2+1 = 3 Marks]**

Space for Rough Works

Space for Rough Works
